

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	B,C	B,D	A,C	A,D	B,D	B,C	A,B
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.00	10.00	64.00	2.50	10.00	11.58		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	C	A,B,D	A,B,C	A,B,C,D	A,C	A	B,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.66 or 1.67	0.00	3.00	5.00	0.77 or 0.78	2.00		

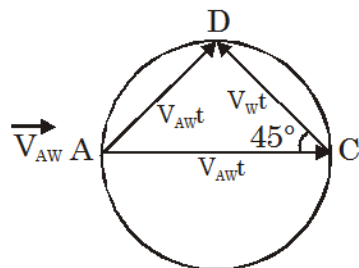
PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	A	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C	A,B,C	A,C	A,C	A	B,C,D	C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.00	0.00	2.66 or 2.67	8.00	3.00	7.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (B)**

(B) Without wind A reaches to C and with wind it reaches to D in same time so wind must deflect from C to D so wind blow in the direction of CD.

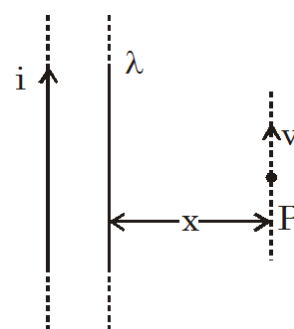


$$\vec{V}_{AG} = \vec{V}_{AW} + \vec{V}_{WG}$$

$$\Rightarrow \vec{V}_{AG}t = \vec{V}_{AW}t + \vec{V}_{WG}t$$

$$AC = \vec{V}_{AW}t$$

$$CD = \vec{V}_{WG}t$$

 2. **Ans (A)**


At P, electric field = $E = \frac{\lambda}{2\pi\epsilon_0 x}$ (to the right),

and magnetic field = $B = \frac{\mu_0 i}{2\pi x}$ (into the paper).

for no deflection, $E = vB$ or $v = \frac{E}{B}$

$$\text{or } v = \frac{\lambda}{2\pi\epsilon_0 x} \times \frac{2\pi x}{\mu_0 i} = \frac{\lambda}{i} \cdot \frac{1}{\epsilon_0 \mu_0} = \frac{\lambda c^2}{i}$$

3. **Ans (B)**

$$\Delta E = 13.6 \times 4 \left(\frac{1}{1} - \frac{1}{9} \right) \text{eV} = 13.6 \times 4 \times \frac{8}{9}$$

$$\text{K.E.}_{\text{max}} = h\nu - \phi$$

$$= [48.35 - 8.35] \text{eV} = 40 \text{eV}$$

$$\lambda_{\text{min}} = \frac{h}{\sqrt{2mK.E.}_{\text{max}}} = \frac{12.27}{\sqrt{V}} \text{\AA} = \frac{12.27}{\sqrt{40}} \text{\AA}$$

$$= 1.94 \text{\AA}$$

4. **Ans (B)**

Constructive interference happens when

$2t = (m - 1/2)\lambda$. The minimum value for m is $m = 1$;

the maximum value is the integer portion of

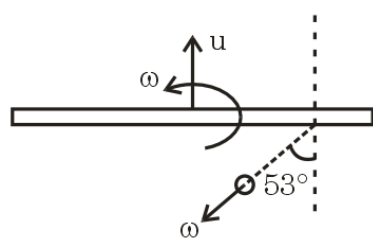
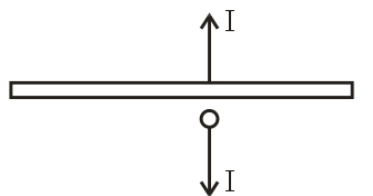
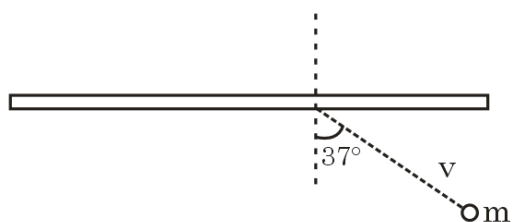
$$\frac{2d}{\lambda} + \frac{1}{2} = \frac{2 \times 0.034 \times 10^{-3}}{680 \times 10^{-9}} + \frac{1}{2} = 100.5$$

$$m_{\text{max}} = 100.$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**



$$mv \sin 37^\circ = m\omega \sin 53^\circ \Rightarrow \omega = \frac{3}{4}v \quad \dots(i)$$

$$I = m[\omega \cos 53^\circ + v \cos 37^\circ] = m \left[\frac{3}{4}v \left(\frac{3}{5} \right) + v \left(\frac{4}{5} \right) \right]$$

$$= \frac{5}{4}mv \quad \dots(2)$$

$$I = (25m)u \Rightarrow u = \frac{v}{20} \quad \dots(3)$$

$$I \left(\frac{\ell}{4} \right) = \left[\frac{(25m)\ell^2}{12} \right] \omega = \frac{5}{16}mv\ell$$

$$\omega = \frac{3v}{20\ell} \quad \dots(3)$$

Heat produced

$$= \frac{1}{2}mv^2 - \left[\frac{1}{2}m \left(\frac{3v}{4} \right)^2 + \frac{1}{2}(25m) \left(\frac{v}{20} \right)^2 + \frac{1}{2} \frac{(25m)\ell^2}{12} \times \left(\frac{3v}{20\ell} \right)^2 \right]$$

$$= \frac{21}{128}mv^2$$

% of K.E. converted into Heat

$$= 32.81\%$$

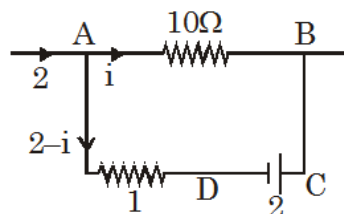
6. **Ans (B,C)**

As $F_1 : F_2 : F_3 = 3 : 5 : 7$, string is fixed at one end, its fundamental frequency is

$$f_0 = \frac{f_1}{3} = \frac{105}{3} = 35 \text{Hz}$$

7. **Ans (B,D)**

Let the currents be as shown in the figure.



KVL along ABCDA

$$-10i - 2 + (2 - i)1 = 0$$

$$i = 0$$

Potential difference across $s = (2 - i)1 = 2 \times 1 = 2\text{V}$.

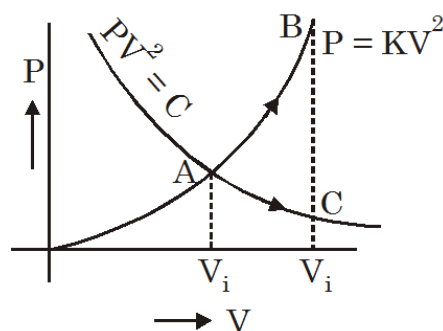
10. **Ans (B,D)**

From equation $PV = nRT$

$$P_C < P_B, V_C = V_B \text{ and } T_B > T_C$$

$$W_{AB} > W_{AC} (T_B - T_A) > T_C - T_A$$

So By 1st law

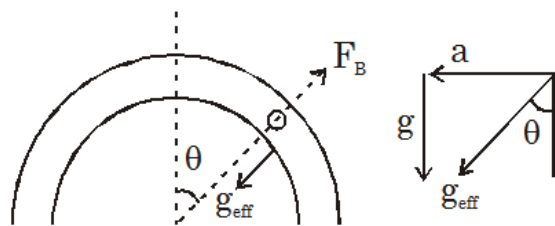


$$Q = \Delta U + W$$

$$Q = nR\Delta T + W$$

$$Q_{AB} > Q_{AC}$$

11. Ans (B,C)



mass of air bubble is neglected

$$\vec{F}_B = -\rho V_d \vec{g}_{\text{eff}}$$

as bubble is towards right, acc. of should also towards right.

$$\tan \theta = \frac{a}{g}$$

12. Ans (A,B)

$$|F| = \frac{dU}{dr} = \frac{Ke^2}{r^4} \quad \dots(1)$$

$$\frac{Ke^2}{r^4} = \frac{mv^2}{r} \quad \dots(2)$$

$$\text{and } mvr = \frac{nh}{2\pi} \quad \dots(3)$$

By (2) and (3)

$$x = \frac{Ke^2 4\pi^2}{h^2} \frac{m}{n^2} = K_1 \frac{m}{n^2} \quad \dots(4)$$

$$\text{Total energy} = \frac{1}{2} (\text{potential energy})$$

$$= \frac{Ke^2}{6r^3} = \frac{-Ke^2}{6 \left(\frac{K_1 m}{n^2} \right)^3} = \frac{-Ke^2 n^6}{6K_1^3 m^3}$$

$$\text{Total energy} \propto n^6$$

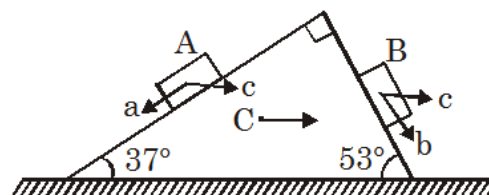
$$\text{Total energy} \propto n^{-3}$$

$\therefore$ (A) and (B) are correct.

PART-1 : PHYSICS

SECTION-II

1. Ans (0.00)



Let c be acceleration of wedge C.

a be acceleration of block A w.r.t. wedge C.

b be acceleration of block B w.r.t. wedge C.

Applying Newton's law in horizontal direction of system of A + B + C.

$$mc + m(c - a \cos 37^\circ) + m(c + b \cos 53^\circ) = 0 \quad \dots(1)$$

Applying Newton's law to block A and B along the incline gives.

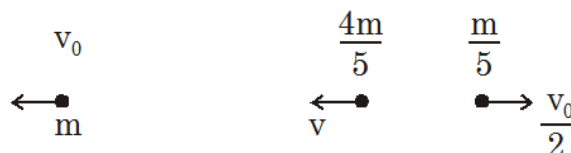
$$\text{In case of A : } mg \sin 37^\circ = m(a - c \cos 37^\circ) \quad \dots(2)$$

$$\text{In case of B : } mg \sin 53^\circ = m(b + c \cos 53^\circ) \quad \dots(3)$$

solving (1), (2) & (3) we get $c = 0$

$$\text{Ans. } a_c = 0$$

3. Ans (64.00)



$$mv_0 = 0.8mv - 0.1mv_0$$

$$v = \frac{11}{8}v_0 = \frac{11}{8} \sqrt{\frac{GM}{4R}} \quad \dots(1)$$

$$E = \frac{1}{2} \left(\frac{4}{5}m \right) v^2 - \frac{GM \left(\frac{4}{5}m \right)}{4R}$$

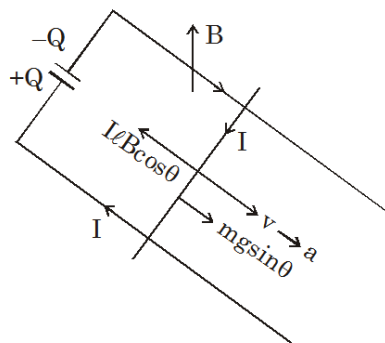
$$= \frac{4}{5}m \left[\frac{121}{512} - \frac{1}{4} \right] \frac{GM}{R} = -GM \left(\frac{\frac{4}{5}m}{2a} \right)$$

$$\Rightarrow \frac{1}{2a} = \left(\frac{1}{4} - \frac{121}{512} \right) \frac{1}{R} = \frac{7}{512R}$$

$$a = \frac{256}{7}R$$

$$\frac{7a}{4R} = 64$$

4. Ans (2.50)



$$mg \sin \theta - IlB \cos \theta = m \frac{dv}{dt} \quad \dots(i)$$

$$vBl \cos \theta = \frac{Q}{C}$$

$$Q = (BlC \cos \theta) = v \quad \dots(ii)$$

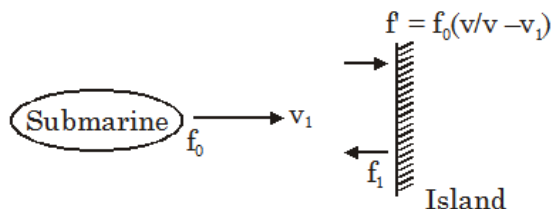
$$I = \frac{d\theta}{dt} = (BlC \cos \theta) \frac{dv}{dt}$$

$$\Rightarrow mg \sin \theta - B^2 l^2 C \cos^2 \theta \frac{dv}{dt} = m \frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{mg \sin \theta}{m + B^2 l^2 C \cos^2 \theta} = \frac{(1)(10) \left(\frac{1}{2}\right)}{1 + \left(\frac{20}{\sqrt{3}}\right)^2 (1)^2 (10^{-2}) \left(\frac{3}{4}\right)}$$

$$= \frac{5}{1+1} = 2.5 \text{ m/s}^2$$

5. Ans (10.00)

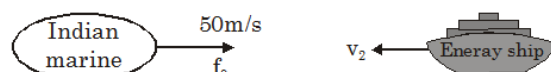


$$f' = f' \left(\frac{v + v_1}{v} \right)$$

$$f' = f \left(\frac{v + v_1}{v - v_1} \right), v = 1050$$

$$\left[f'' = f + \frac{10f}{100} \right]$$

$$v_1 = 50 \text{ m/sec.}$$



$$f' = f' \left(\frac{v + 50}{v - v_2} \right)$$

$$f_1 = f_0 \left(\frac{v + v_2}{v - 50} \right)$$

$$f' = f_0 \left(\frac{(v + v_2)(v + 50)}{(v - v_2)(v - 50)} \right) = 1.21f_0$$

[21% greater than sent waves]

get $v_2 = 50 \text{ m/sec}$ toward Indian submarine.

6. Ans (11.58)

$$\tan \phi = \left| \frac{X_C - X_L}{R} \right|$$

$$\tan 37^\circ = \frac{X_L}{120\Omega} \Rightarrow X_L = 90\Omega$$

$$\tan 53^\circ = \frac{X_C}{120\Omega} \Rightarrow X_C = 160\Omega$$

$$z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{(120)^2 + (70)^2}$$

$$= 10\sqrt{193}\Omega$$

$$P = \frac{V_0^2}{2Z^2} R = \frac{(193)^2}{2[100 \times 193]} \times 120 = \frac{193 \times 60}{100} \text{ W}$$

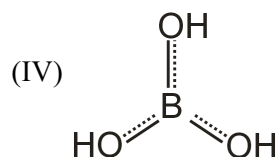
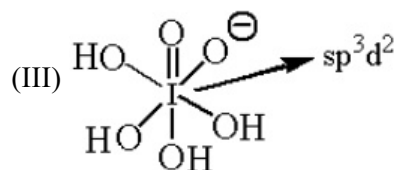
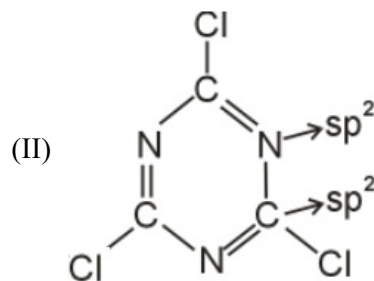
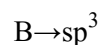
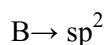
$$P = 115.8 \text{ W}$$

$$\frac{P}{10} = 11.58$$

PART-2 : CHEMISTRY

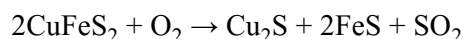
SECTION-I (i)

1. Ans (C)

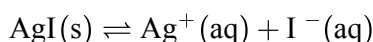


2. Ans (D)

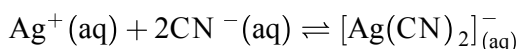
The reaction at high temperature in the blast furnace is :



4. **Ans (B)**



$$K_{sp} = [\text{Ag}^+(\text{aq})][\text{I}^-(\text{aq})]$$



$$K_f = \frac{[\text{Ag}(\text{CN})_2]^{-}}{[\text{Ag}^+][\text{CN}^-]^2}$$

Now let a moles of AgI dissolved in CN^- solution.



$$1 \text{ mole} \quad 0 \quad 0$$

$$(1 - 2x) \text{ mole} \quad x \text{ mole} \quad x \text{ mole}$$

$$\text{As } [\text{CN}^-] = MV = 1 \times 1 = 1 \text{ mole}$$

$$K_c = \frac{x \cdot x}{(1 - 2x)^2}$$

$$\text{also } K_c = K_f K_{sp}$$

$$\frac{x^2}{(1 - 2x)^2} = 1.19 \times 10^{-17} \text{M}^2 \times 7.11 \times 10^{19} \text{M}^{-2}$$

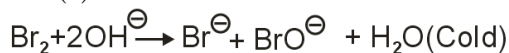
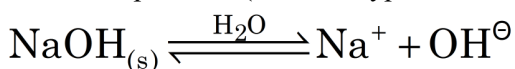
$$x = 0.5$$

PART-2 : CHEMISTRY

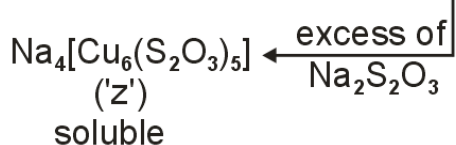
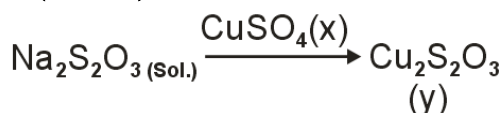
SECTION-I (ii)

5. **Ans (C)**

Bromides and iodides do not interfere with the chromyl chloride test, because they are oxidised to Br_2 (red vapour) and I_2 (violet vapours) which dissolve in the NaOH solution to form colourless products (halide + hypohalite/halate)



6. **Ans (A,B,D)**



7. **Ans (A,B,C)**

∴ (A) Etard reaction

(B) Mild oxidising agent

(C) Rosenmund reduction

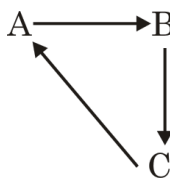
(D) Haloform reaction

8. **Ans (A,B,C,D)**

∴ All reagents will convert it into alkane.

10. **Ans (A)**

Over all process is cyclic



$$\text{Thus, } \Delta E = 0, \Delta H = 0, \Delta S = 0, \Delta E = q + W$$

$$0 = q + W \Rightarrow q = -W$$

$$W = -\frac{1}{2} \times (40 - 20)(1 - 0.5)$$

$$= -5 \text{ atm L}$$

$$= -5 \times 101.3 \text{ J}$$

$$= -506.5 \text{ J}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (1.66 or 1.67)**

$$x = [\text{Co}(\text{NH}_3)_2(\text{en})_2]^{+3}, [\text{Co}(\text{NH}_3)(\text{Br})(\text{en})_2]^{+2}, [\text{Co}(\text{gly})_3]^0$$

$$y = [\text{Co}(\text{NH}_3)_2(\text{en})_2]^{+3}, [\text{Co}(\text{NH}_3)(\text{Br})(\text{en})_2]^{+2}, [\text{Zn}(\text{gly})_2]^0, [\text{Fe}(\text{EDTA})]^\oplus, [\text{Co}(\text{gly})_3]^0$$

$$\frac{y}{x} \Rightarrow \frac{5}{3} = 1.66 \text{ or } 1.67$$

2. Ans (0.00)

$A \Rightarrow \text{Pb}_2\text{O}_3 \Rightarrow \text{PbO} \cdot \text{PbO}_2$ Reddish - yellow

$\text{Pb}_3\text{O}_4 \Rightarrow 2\text{PbO} \cdot \text{PbO}_2$ Red

$B \Rightarrow \text{PbO}_2$ dark brown

$\text{PbO} \Rightarrow$ yellow - massicot

red or reddish-yellow - litharge

$C \Rightarrow \text{PbO}_2$

$\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$

$\text{Pb}_2\text{O}_3 + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 \downarrow + \text{H}_2\text{O}$

Brown ppt.

$\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \xrightarrow[\text{cold conc.}]{\text{hot dil. or}} 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + \text{H}_2\text{O}$

$A \Rightarrow 2$

$B \Rightarrow 1$

$C \Rightarrow 1$

$D \Rightarrow 0$

Ans. 0.00

5. Ans (0.77 or 0.78)

$P_B^\circ = 40$ $P_T^\circ = 20$ $K_B = 0.5 = K_T$

$$\begin{aligned} \text{Now } y_B &= \frac{K_B P_B^\circ}{K_B P_B^\circ + K_T P_T^\circ} \\ &= \frac{70 \times 0.5}{70 \times 0.5 + 20 \times 0.5} \\ &= 0.777 \end{aligned}$$

6. Ans (2.00)

$$\lambda = \frac{h}{\sqrt{2m_p eV}} \quad M_\alpha = 4M_p$$

$$\lambda_2 = \frac{h}{\sqrt{2M_\alpha 2eV}} \quad q_\alpha = 2e$$

$$\lambda_2 = \frac{h}{\sqrt{2M_p 8eV_2}}$$

if $\lambda_2 = \lambda$ then $8V_2 = 16$

$$V_2 = \frac{16}{8}$$

$V_2 = 2$ volt

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

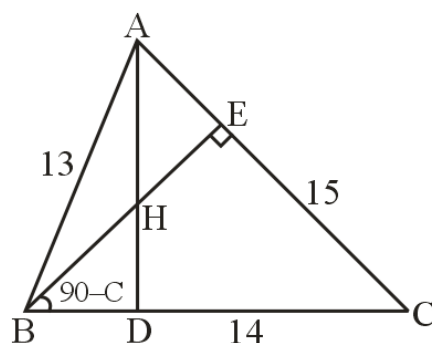
$$(\sqrt{5} + \sqrt{3})^{100} = I + f$$

$$(\sqrt{5} - \sqrt{3})^{100} = f'$$

$$I + f + f' = 2 ({}^{100}C_0 5^{50} + {}^{100}C_2 5^{49} + \dots)$$

$$\Rightarrow f + f' = 1$$

2. Ans (A)



$$\sin(90^\circ - C) = \frac{HD}{HB}$$

$$\Rightarrow \frac{HD}{HB} = \cos C$$

$$= \frac{14^2 + 15^2 - 13^2}{2 \cdot 14 \cdot 15}$$

$$= \frac{3}{5}$$

3. Ans (D)

$$z = \frac{(\sqrt{3} - i)^{100}}{4^{100}} = \frac{2^{100}}{4^{100}} e^{-i \frac{\pi}{6} \times 100}$$

$$= \frac{1}{2^{100}} e^{-\left(16\pi + \frac{4\pi}{6}\right)i}$$

$$= \frac{1}{2^{100}} e^{-\frac{2\pi}{3}i}$$

4. **Ans (B)**

$$\begin{aligned} 1 + \sin^2 x \theta &= \cos \theta \\ \Rightarrow (1 - \cos \theta) + \sin^2 x \theta &= 0 \\ \Rightarrow 2\sin^2(\theta/2) + \sin^2 x \theta &= 0 \\ \Rightarrow \sin(\theta/2) = 0 \text{ and } \sin x \theta &= 0 \\ \Rightarrow \theta = 2n\pi \text{ and } \theta = m\pi/x \\ \Rightarrow m\pi/x = 2n\pi \\ \Rightarrow x &= m/2n \end{aligned}$$

Option (B) is correct

PART-3 : MATHEMATICS

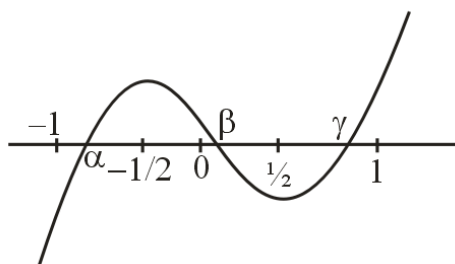
SECTION-I (ii)

5. **Ans (B,C)**

$6561 = 3^8$ and T_n of A.P $= 1 + (n-1)3 = 3n - 2$
number immediately preceding $= T_7 = 19$
number immediately succeeding $= T_8 = 22$

6. **Ans (A,B,C)**

$$\begin{aligned} f(x) &= 8x^3 - 6x + 1 \\ f(x) &= 6(4x^2 - 1) \end{aligned}$$



$$\text{Also } 3x - 4x^3 = \frac{1}{2}$$

$$\text{put } x = \sin \theta$$

$$\sin 3\theta = \frac{1}{2}$$

$$\Rightarrow 3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \dots$$

$$\Rightarrow \alpha = \sin \frac{25\pi}{18}, \beta = \sin \frac{\pi}{18}, \gamma = \sin \frac{5\pi}{18} \text{ or}$$

$$\sin \frac{13\pi}{18}$$

7. **Ans (A,C)**

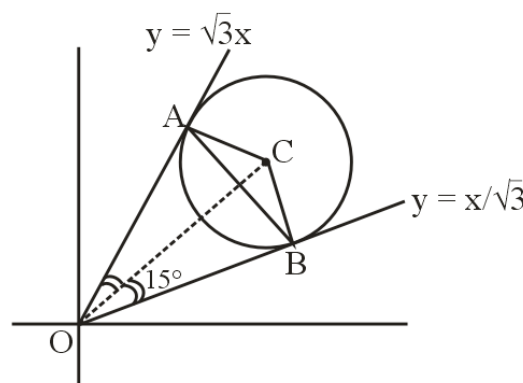
$$\text{roots are } -1 \text{ and } -\left(\frac{b+c-a}{a+c-b}\right)$$

$$\frac{b+c-a}{a+c-b} = 2 \Rightarrow 3b = 3a + c$$

OR

$$\frac{2(b+c-a)}{a+c-b} = 1 \Rightarrow 3b - 3a = -c$$

8. **Ans (A,C)**



centre lies on line $y = x$

$$\sin 15^\circ = \frac{r}{OC} \Rightarrow OC = \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} 2\sqrt{2}$$

$$OC = 2\sqrt{2}$$

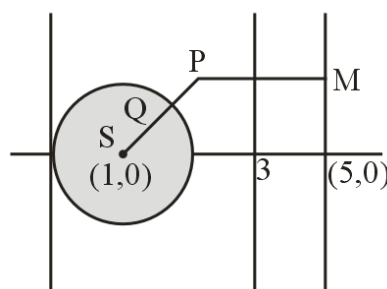
$$\Rightarrow C(2, 2)$$

circumcircle of $\triangle OAB$ is $x(x-2) + y(y-2) = 0$

$$R = \sqrt{2}$$

$$\frac{AB}{\sin 30^\circ} = 2R \Rightarrow AB = 2\sqrt{2} \cdot \frac{1}{2} = \sqrt{2}$$

9. **Ans (A)**



$$\frac{PQ}{PM} = \frac{1}{2} \text{ where } PM > 2$$

$$\frac{PS-1}{PM} = \frac{1}{2}$$

$$\frac{PS}{PM} = \frac{1}{2} + \frac{1}{PM} < 1$$

10. **Ans (B,C,D)**

$$\text{Put } \cos^{-1} \sqrt{x} = \theta \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow f(x) = 2\sin^{-1} \sin \theta + \sin^{-1} \sin 2\theta$$

$$= \begin{cases} 4\theta & 0 \leq \theta \leq \frac{\pi}{4} \\ 2\theta + \pi - 2\theta & \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2} \end{cases}$$

$$= \begin{cases} 4\cos^{-1} \sqrt{x} & \frac{1}{2} \leq x \leq 1 \\ \pi & 0 \leq x \leq \frac{1}{2} \end{cases}$$

11. Ans (C,D)

$$F'(x) = f(x) (1 + 2f(x) + 3f^2(x))$$

$f(x)$ is continuous & strictly increasing so $f'(x) > 0$

$$\text{and } (1 + 2f(x) + 3f^2(x)) > 0 (\because D < 0)$$

So $F'(x) > 0 \Rightarrow F(x)$ is increasing function.

if $F(x)$ has positive root then $F(0) < 0$

$$F(0) = (1 + f(0)) \underbrace{(1 + f^2(0))}_{+ve} < 0$$

$$1 + f(0) < 0$$

12. Ans (B,C)

$$\frac{2xy' - x^2y''}{(y')^2} = x$$

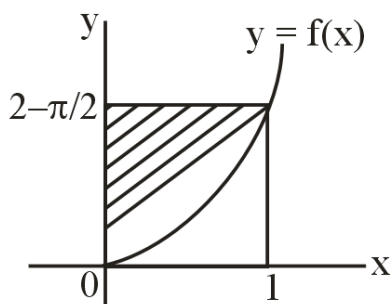
$$\Rightarrow \int d\left(\frac{x^2}{y}\right) = \int x dx$$

$$\Rightarrow \frac{x^2}{y} = \frac{x^2}{2} + c \text{ or } y' = \frac{2x^2}{x^2 + k}$$

$$\text{given } f''(1) = 1 \Rightarrow k = 1$$

$$\Rightarrow y = \int \frac{2x^2}{x^2 + 1} dx = 2x - 2\tan^{-1}x$$

$f(x)$ is strictly increasing



$$A = \left(2 - \frac{\pi}{2}\right) - \int_0^1 (2x - 2\tan^{-1}x) dx$$

$$= \left(2 - \frac{\pi}{2}\right) - (x^2 - 2x\tan^{-1}x + \ln(1+x^2)) \Big|_0^1$$

$$= 2 - \frac{\pi}{2} - \left(1 - \frac{\pi}{2} + \ln 2\right) = 1 - \ln 2$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3.00)

Any point on the line $x + y = 1$ can be taken

$$(t, 1 - t)$$

Equation of the chord, with this as mid point is

$$y(1 - t) - 2a(x + t) = (1 - t)^2 - 4at$$

It passes through $(a, 2a)$

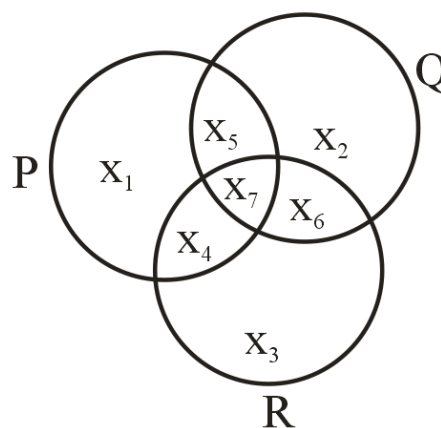
$$\text{So, } t^2 - 2t + 2a^2 - 2a + 1 = 0, \text{ this should have}$$

2 distinct real roots

$$\text{so } a^2 - a < 0, 0 < a < 1$$

$$\Rightarrow \text{length of latus rectum} < 4$$

2. Ans (0.00)



Let 'a' number of people read no newspaper at all.

$$\text{Now, } x_1 + x_4 + x_5 + x_7 = 500$$

$$x_2 + x_5 + x_6 + x_7 = 500$$

$$x_3 + x_4 + x_6 + x_7 = 500$$

$$x_5 + x_7 = 250$$

$$x_4 + x_7 = 250$$

$$x_6 + x_7 = 250$$

$$x_7 = 250$$

$$\text{And, } x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + a = 1000$$

$$\Rightarrow x_4 = x_5 = x_6 = 0$$

$$\Rightarrow x_1 = 250, x_2 = 250, x_3 = 250$$

$$\Rightarrow a = 1000 - 250 - 250 - 0 - 0 - 0 - 250 = 0$$

3. Ans (2.66 or 2.67)

Let the height is h and width is x

Therefore, length = $2x$

$$\text{Volume} = x \cdot 2x \cdot h = 2x^2h = 48$$

$$\Rightarrow x^2 = 24/h$$

Cost = $c \cdot x \cdot 2x + 3c(x \cdot 2x + 2 \cdot x \cdot h + 2 \cdot 2x \cdot h)$ where c is cost per sq ft

$$\text{Cost} = C = c(48/h) + 3c(48/h + 6h\sqrt{(24/h)})$$

$$= 4.48c/h + 36c\sqrt{(6h)}$$

$$dC/dh = -4.48c/h^2 + 36c\sqrt{(6/2)}\sqrt{h} = 0$$

$$\Rightarrow h^{3/2} = 4.48 \cdot 2/36\sqrt{6} = 16 \cdot \sqrt{2/3}^{3/2}$$

$$\Rightarrow h^{3/2} = (8/3)^{3/2}$$

$$\Rightarrow h = 8/3$$

4. Ans (8.00)

$$\text{tr}(A) = (x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 = 0$$

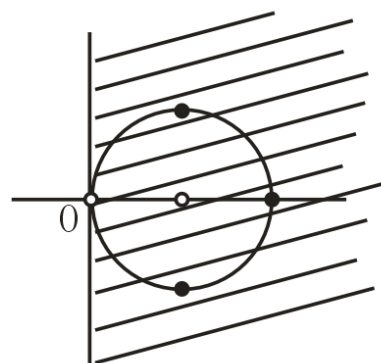
$$|A| = \begin{vmatrix} x_1^2 & -2x_1 & 1 \\ x_2^2 & -2x_2 & 1 \\ x_3^2 & -2x_3 & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ y_1 & y_2 & y_3 \\ y_1^2 & y_2^2 & y_3^2 \end{vmatrix}$$

$$= 2(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)$$

$$= 2(x_1 - x_2)^2(x_2 - x_3)^2(x_3 - x_1)^2 \geq 8$$

when $x_1 = 1$, $x_2 = 2$ and $x_3 = 3$

5. Ans (3.00)



$$\text{Re}\left(\frac{1}{x + iy}\right) \geq \frac{1}{2}$$

$$\text{Re}\left(\frac{x - iy}{x^2 + y^2}\right) \geq \frac{1}{2}$$

$$\Rightarrow 2x \geq x^2 + y^2$$

$$\therefore z \neq 0, 1$$

$\therefore$ 3 integral points.

6. Ans (7.00)

Let $T \equiv$ Test positive, $S \equiv$ sufferer

$$P\left(\frac{T}{S}\right) = .9, P\left(\frac{T}{\bar{S}}\right) = .10, P(S) = \frac{1}{100} = .01$$

$$P\left(\frac{S}{T}\right) = \frac{P\left(\frac{T}{S}\right)P(S)}{P\left(\frac{T}{S}\right)P(S) + P\left(\frac{T}{\bar{S}}\right)P(\bar{S})}$$

$$= \frac{.9 \times .01}{.9 \times .01 + .1 \times .99}$$

$$= \frac{9}{9 + 99} = \frac{9}{108} = \frac{1}{12}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	C,D	A,B,C	A,C	C,D	C	A,D	A,C,D	A,B
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	40.00	0.33 to 0.34	2.00	10.00	0.16		

PART-2 : CHEMISTRY

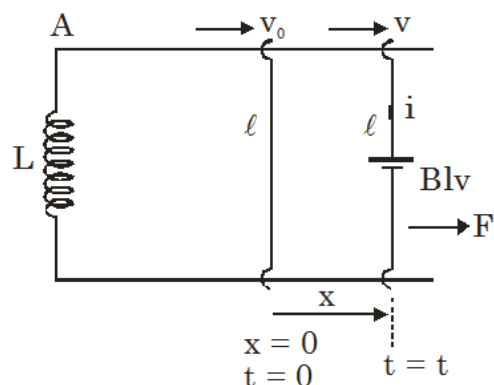
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B	A,B,C	B,C	A,B,C	A,B,C,D	A,C,D	A,B,D	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	B	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	5.00	7.00	4.00	3.20	0.50	1.33		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C	A,B	B,D	A,B,C	A,B,C,D	A,B,C	A,B,C	A,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	B	A	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	2.00	10.00	8.00	20.00	2.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (C,D)



$$V_A - L \frac{di}{dt} + B\ell v = V_A$$

$$L \frac{di}{dt} = B\ell v$$

$$L \int_0^i di = B\ell \int_0^x dx$$

$$Li = B\ell x$$

$$F_{\text{net}} = -B\ell i$$

$$ma = -B \cdot \left(\frac{B\ell x}{L} \right) \ell$$

$$a = - \left(\frac{B^2 \ell^2}{mL} \right) x$$

$$w = \frac{B\ell}{\sqrt{mL}}$$

$$T = \frac{2\pi}{w} = \frac{2\pi\sqrt{mL}}{B\ell}$$

$$= \frac{2\pi \times \sqrt{80 \times 10^{-3} \times 2}}{4 \times 4 \times 10^{-2}}$$

$$= \frac{2\pi \times 4 \times 10^{-1}}{4 \times 4 \times 10^{-2}}$$

$$= \frac{\pi}{2} \times 10$$

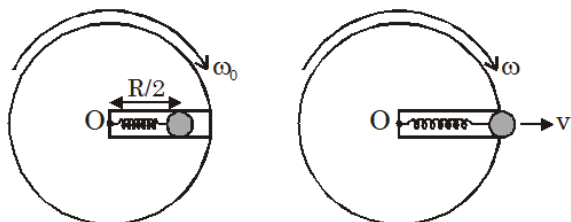
$$T = 5\pi \text{ sec}$$

$$V_0 = A\omega$$

$$2 = A \times \left(\frac{2\pi}{5\pi} \right)$$

$$A = 5\text{cm}$$

2. Ans (A,B,C)



Angular momentum conservation about 'O'

$$\Rightarrow \left(\frac{mR^2}{2} + \frac{mR^2}{4} \right) \omega_0 = \left(\frac{mR^2}{2} + mR^2 \right) \omega$$

$$\omega = \frac{\omega_0}{2}$$

From energy conservation

$$\frac{1}{2} \left(\frac{mR^2}{2} + \frac{mR^2}{4} \right) \omega_0^2 + 0$$

$$= \frac{1}{2} \left(\frac{mR^2}{2} + mR^2 \right) \left(\frac{\omega_0}{2} \right)^2 + \frac{1}{2}mv^2 + \frac{1}{2}K \left(\frac{R}{2} \right)^2$$

$$\frac{3}{8}mR^2\omega_0^2 = \frac{3}{16}mR^2\omega_0^2 + \frac{1}{2}mv^2 + \frac{1}{2} \frac{m\omega_0^2}{2} \cdot \frac{R^2}{4}$$

$$\frac{3}{16}mR^2\omega_0^2 - \frac{1}{16}mR^2\omega_0^2 = \frac{1}{2}mv^2 \Rightarrow v = \frac{\omega_0 R}{2}$$

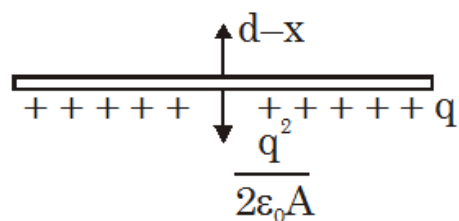
$$\text{Energy of spring } E = \frac{1}{2}Kx^2$$

$$\frac{dE}{dt} = \frac{1}{2} \cdot K \cdot 2x \cdot \frac{dx}{dt} = Kxv$$

$$= \frac{m\omega_0^2}{2} \cdot \frac{R}{2} \cdot \frac{\omega_0 R}{2}$$

$$= \frac{m\omega_0^3 \cdot R^2}{8}$$

3. Ans (A,C)



$$K(d-x) = \frac{q^2}{2\epsilon_0 A}$$

$$K(d-x) = \frac{(CV)^2}{2\epsilon_0 A}$$

$$V = \sqrt{\frac{2\epsilon_0 AK \cdot (d-x)}{C^2}}$$

$$= \sqrt{\frac{2\epsilon_0 AK \cdot (d-x)}{\left(\frac{\epsilon_0 A}{x} \right)^2}}$$

$$V = \sqrt{\frac{2Kx^2(d-x)}{\epsilon_0 A}}$$

$$\frac{dV}{dx} = 0$$

$$2x \cdot d - 3x^2 = 0$$

$$x = \frac{2d}{3}$$

$$V_{\max} = \sqrt{\frac{2K}{\epsilon_0 A} \cdot \frac{4d^2}{9} \cdot \frac{d}{3}} = \sqrt{\frac{8Kd^3}{27\epsilon_0 A}}$$

4. **Ans (C,D)**

$$f = -Kv$$

$$m \cdot \frac{dv}{dt} = -Kv$$

$$\int_{v_0}^v \frac{dv}{v} = -\frac{K}{m} \int_0^t dt$$

$$\ln\left(\frac{v}{v_0}\right) = -\frac{K}{m} \cdot t$$

$$v = v_0 e^{-\frac{K}{m} \cdot t}$$

$$R = \frac{mv}{qB} = \frac{mv_0}{qB} e^{-\frac{K}{m} \cdot t}$$

$$\text{From } f = -Kv$$

$$m \cdot v \frac{dv}{dx} = -Kv$$

$$\int_{v_0}^v dv = -\frac{K}{m} \cdot \int_0^x dx$$

$$v = v_0 - \frac{K}{m} \cdot x = v_0 - \frac{K}{m} \cdot \frac{mv_0}{2K}$$

$$v = \frac{v_0}{2}$$

$$R = \frac{mv}{qB} = \frac{mv_0}{2qB}$$

5. **Ans (C)**

$$\text{For } \lambda = 600\text{nm}$$

$$E = \frac{hc}{\lambda} = \frac{1240}{600} = 2.067\text{eV}$$

$$\text{work function } \phi = 2.39\text{eV}$$

$$\text{For } \lambda = 400\text{nm}, E = \frac{1240}{400}$$

$$= 3.1\text{ eV}$$

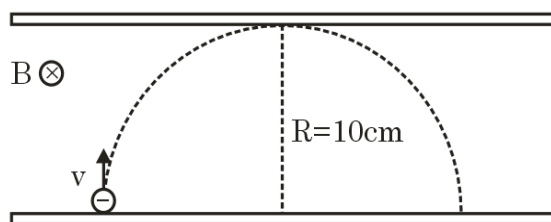
$$E > \phi \Rightarrow \text{There is photo emission.}$$

$$\text{From } E = K_{\max} + \phi$$

$$\frac{hc}{\lambda} = K_{\max} + \phi$$

$$\frac{1240}{400} = K_{\max} + 2.39$$

$$K_{\max} = 0.71\text{eV}$$



$$R = \frac{mv}{qB} = \frac{\sqrt{2 \cdot K_{\max} \cdot m}}{qB}$$

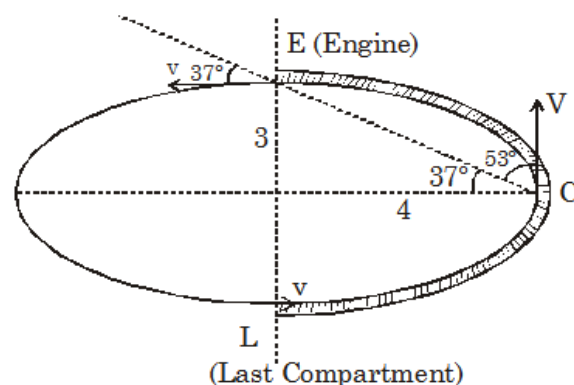
$$R = \frac{\sqrt{2 \times 0.71 \times 1.6 \times 10^{-19} \times 9 \times 10^{-31}}}{1.6 \times 10^{-19} \times B}$$

$$B = \frac{1}{R} \sqrt{\frac{2 \times 0.71 \times 9 \times 10^{-31}}{1.6 \times 10^{-19}}}$$

$$= \frac{1}{10 \times 10^{-2}} \sqrt{\frac{2 \times .71 \times 9}{1.6} \times 10^{-12}}$$

$$B = 2.826 \times 10^{-5} \text{T}$$

6. **Ans (A,D)**



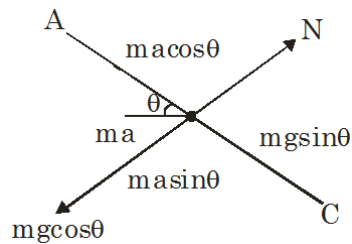
$$\text{Frequency received by L is } = 3460 \text{ Hz}$$

$$\text{Frequency received by C is } f' = f \left[\frac{C + v \cos 53^\circ}{C + v \cos 37^\circ} \right]$$

$$= 3460 \left[\frac{330 + 20 \times 0.6}{330 + 20 \times 0.8} \right]$$

$$= 3420\text{Hz}$$

7. Ans (A,C,D)



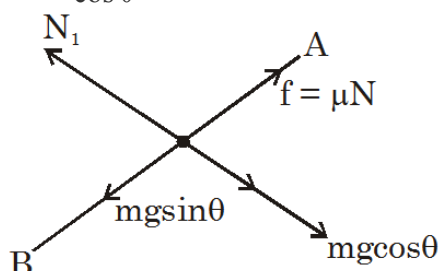
$$ma \cos \theta = mg \sin \theta$$

$$a = g \tan \theta$$

$$N = mg \cos \theta + ma \sin \theta$$

$$N = mg \cos \theta + m \cdot g \tan \theta \cdot \sin \theta$$

$$N = \frac{mg}{\cos \theta}$$



$$N_1 = mg \cos \theta$$

$$N_2 = ma$$

$$= mg \tan \theta$$

$$N = \sqrt{N_1^2 + N_2^2}$$

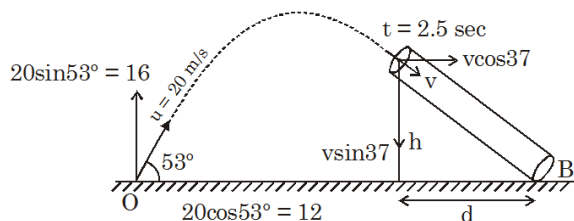
$$N = mg \sqrt{\cos^2 \theta + \tan^2 \theta}$$

$$\mu N = mg \sin \theta$$

$$\mu \cdot mg \sqrt{\cos^2 \theta + \tan^2 \theta} = mg \sin \theta$$

$$\mu = \frac{\sin \theta}{\sqrt{\cos^2 \theta + \tan^2 \theta}}$$

8. Ans (A,B)



$$v \cos 37 = 20 \cos 53$$

$$v = 20 \times \frac{3}{4} = 15 \text{ m/s}$$

$$v_y = u_y + a_y t$$

$$-v \sin 37 = 16 - 10t$$

$$-15 \times \frac{3}{5} = 16 - 10t$$

$$t = 2.5 \text{ sec}$$

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$h = 16 \times 2.5 + \frac{1}{2} \times (-10) \times 2.5^2$$

$$= 40 - \frac{62.5}{2}$$

$$= 40 - 31.25$$

$$h = 8.75 \text{ m}$$

$$\sin 37 = \frac{h}{L}$$

$$L = \frac{h}{\sin 37} = \frac{8.75}{\frac{3}{5}} = \frac{875 \times 5}{300} = \frac{175}{12} \text{ m}$$

$$x = u_x t$$

$$= 12 \times 2.5$$

$$= 30 \text{ m}$$

$$d = L \cos 37$$

$$d = \frac{175}{12} \times \frac{4}{5}$$

$$= \frac{35}{3}$$

$$OB = 30 + \frac{35}{3} = \frac{125}{3} \text{ m}$$

$$L = vt + \frac{1}{2} g \sin 37^\circ t^2$$

$$\frac{175}{12} = 15t + 3t^2$$

$$36t^2 + 180t - 175 = 0$$

$$t_{OB} = 2.5 + t$$

PART-1 : PHYSICS

SECTION-I (ii)

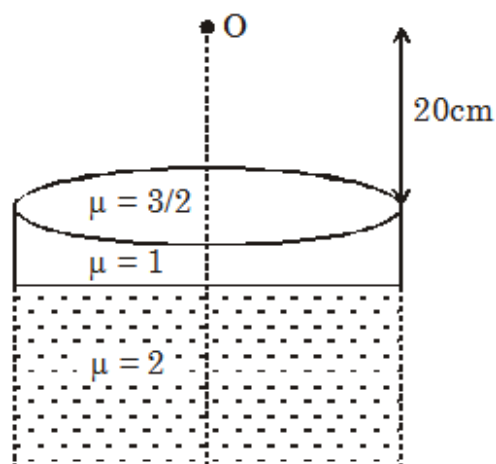
12. Ans (C)

By using this equation $m = \frac{v\mu_1}{u\mu_2}$, we get the correct combination.

PART-1 : PHYSICS

SECTION-II

2. Ans (40.00)



Power

$$P = \left(\frac{\frac{3}{2} - 1}{R} \right) + \left(\frac{1 - \frac{3}{2}}{-R} \right) + \left(\frac{2 - 1}{\infty} \right)$$

$$P = \frac{1}{2R} + \frac{1}{2R} = \frac{1}{R} \dots (1)$$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{-R} \right)$$

$$\frac{1}{10} = \left(\frac{3}{2} - 1 \right) \left(\frac{2}{R} \right)$$

$$R = 10 \text{ cm}$$

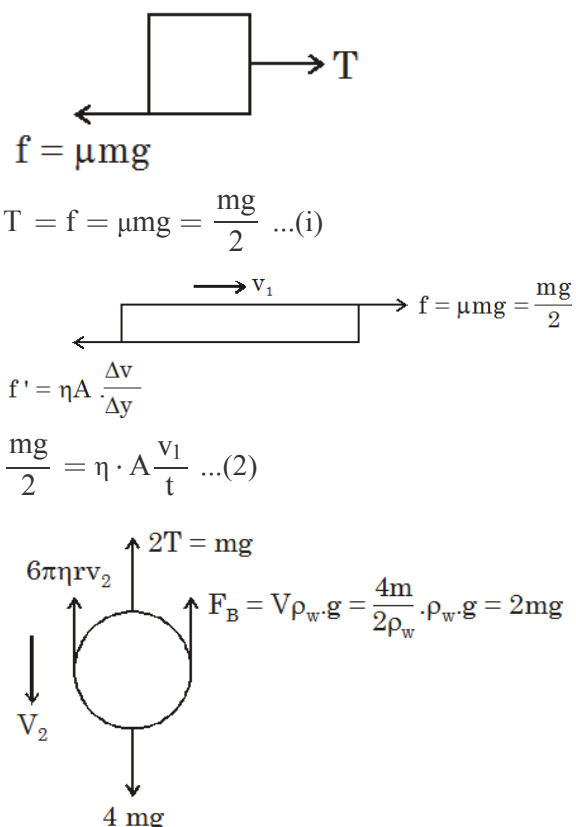
$$\text{From (1)} \Rightarrow P = \frac{1}{10}$$

$$\frac{\mu_2}{v} - \frac{\mu_1}{4} = \frac{1}{10}$$

$$\frac{2}{v} - \frac{1}{-20} = \frac{1}{10}$$

$$v = 40 \text{ cm}$$

3. Ans (0.33 to 0.34)



$$4mg - mg - 2mg - 6πηrv_2 = 0 \dots (3)$$

$$6πηrv_2 = mg$$

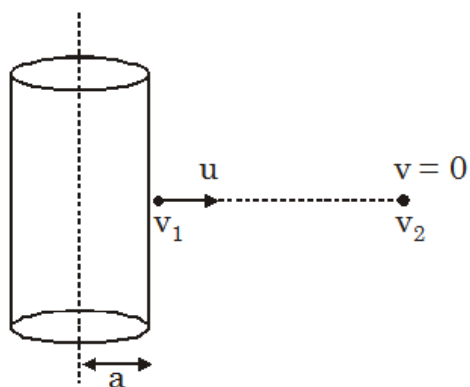
$$\text{From } \frac{3}{2}$$

$$\Rightarrow \frac{6πηrv_2}{\eta A \cdot \frac{v_1}{t}} = 2$$

$$\frac{v_2}{v_1} = \frac{2A}{6\pi r t}$$

$$\Rightarrow N = \frac{2}{6} = \frac{1}{3}$$

4. Ans (2.00)



From energy conservation

$$\frac{1}{2}mu^2 + (-e)v_1 = 0 + (-e)v_2$$

$$u = \sqrt{\frac{2e}{m}(v_1 - v_2)} \dots (1)$$

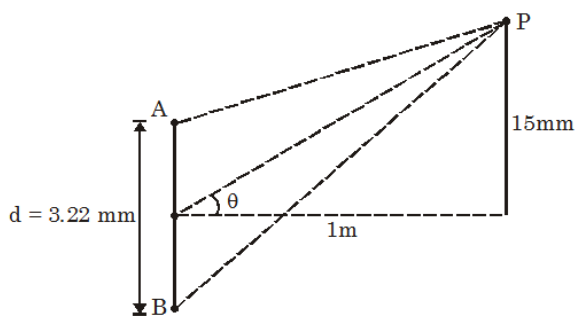
$$v_1 - v_2 = \frac{\sigma a}{\epsilon_0} \cdot \ln\left(\frac{r}{a}\right)$$

From (1)

$$\Rightarrow u = \sqrt{\frac{2e\sigma a}{m\epsilon_0} \ln\left(\frac{r}{a}\right)}$$

$$N = 2$$

5. Ans (10.00)



$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = \frac{2\pi}{\lambda} \cdot d \sin \theta$$

$$\Delta\phi = \frac{2\pi}{600 \times 10^{-9}} \times 3.22 \times 10^{-3} \times \frac{15 \times 10^{-3}}{1}$$

$$\Delta\phi = 161\pi = \pi \Rightarrow \text{Phase diff. b/w A \& B at}$$

point A.

Resultant intensity at point P is = I_C

$$= 10 \text{ mw/m}^2$$

6. Ans (0.16)

$$P = KA \cdot \frac{\Delta\theta}{\Delta x} + F\alpha + nC_V \cdot \frac{dT}{dt} \dots (1)$$

$$P = 4440 \text{ watt.}$$

$$KA \cdot \frac{\Delta\theta}{\Delta x} = K \cdot (2\pi rh) \cdot \frac{\Delta\theta}{\Delta x}$$

$$= 2 \times 2\pi \times \frac{10}{\sqrt{\pi}} \times 10^{-2} \times 50 \times 10^{-2} \times \frac{(20-0)}{\frac{\sqrt{\pi}}{10} \times 10^{-2}}$$

$$= 4000$$

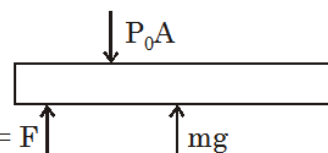
$$n \cdot C_V \cdot \frac{dT}{dt} = n \cdot \frac{3}{2}R \cdot \frac{dT}{dt} = \frac{3}{2} \frac{d}{dt}(nRT)$$

$$= \frac{3}{2} \frac{d}{dt}(Pv)$$

$$= \frac{3}{2} \frac{d}{dt}(P \cdot A \cdot h)$$

$$= \frac{3}{2} \cdot PA \cdot \alpha, \frac{dh}{dt} = \alpha$$

$$= \frac{3}{2} \times 1100\alpha$$



$$F = P_0A + mg$$

$$= 10^5 \times \pi \left(\frac{10}{\sqrt{\pi}} \times 10^{-2} \right)^2 + 100$$

$$= 1100$$

From (1)

$$\Rightarrow 4440 = 4000 + 1100\alpha + \frac{3}{2} \times 1100\alpha$$

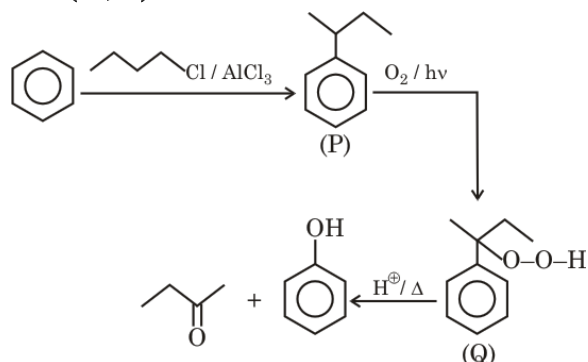
$$440 = \frac{5}{2} \times 1100\alpha$$

$$\alpha = \frac{8}{50} = 0.16 \text{ m/sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,B)



5. Ans (A,B,C,D)

$$V_f=1 \quad V_f=1$$



$$\downarrow \quad \downarrow$$

$$x \text{ mmole} \quad y \text{ mmole}$$

At HPh end point,

$$\text{eq. of HCl} = \text{eq. of Na}_2\text{CO}_3$$

$$\frac{1}{5} \times 2.5 \times 10^{-3} \times 1$$

$$= x \times 10^{-3} \times 1$$

$$\Rightarrow x = 0.5 \text{ mmole} = 0.5 \times 10^6 \text{ mg} = 53 \text{ mg} = 0.053 \text{ g}$$

At Mo end point

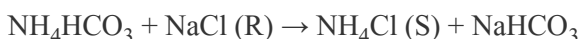
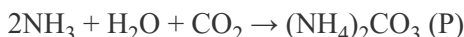
$$\text{eq. of HCl} = 2 \times \text{eq. of Na}_2\text{CO}_3 + \text{eq. of NaHCO}_3$$

$$\frac{1}{5} \times 7.5 \times 10^{-3} \times 1 = (x \times 10^{-3} \times 1) \times 2 + y \times 10^{-3} \times 1$$

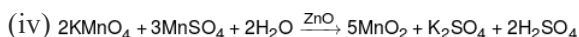
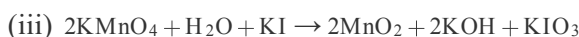
$$\Rightarrow 1.5 = 1 + y \Rightarrow y = 0.5 \text{ mmole}$$

$$= 0.5 \times 84 = 42 \text{ mg} = 0.042 \text{ mg}$$

7. Ans (A,B,D)



8. Ans (B,C,D)



PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (D)

$$\text{By definition, } \alpha = \frac{C_{A_0} - C_A}{C_{A_0}}$$

$$\text{For zero order, } C_A = C_{A_0} - kt \Rightarrow \alpha = \frac{kt}{C_{A_0}}$$

$$\text{For 1st order, } C_A = C_{A_0} e^{-kt}$$

$$\Rightarrow C_{A_0} (1 - \alpha) = C_{A_0} e^{-kt}$$

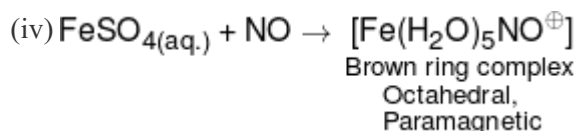
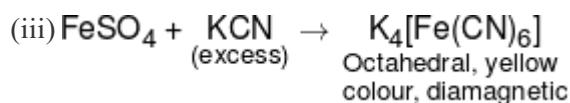
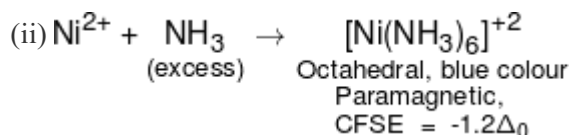
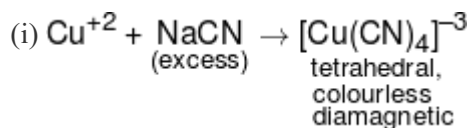
$$\Rightarrow \alpha = 1 - e^{-kt}$$

$$\text{For 2nd order, } \frac{1}{C_A} = \frac{1}{C_{A_0}} + kt$$

$$\text{or, } \frac{C_{A_0} - C_A}{C_A C_{A_0}} = kt$$

$$\text{or, } \alpha = \frac{C_{A_0} kt}{1 + C_{A_0} kt}$$

11. Ans (B)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (5.00)

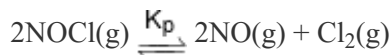
(i), (iii), (iv), (v), (vi)

2. Ans (7.00)

Carbyl amine reaction gives foul smell with 1° amine and aniline.

4. Ans (3.20)

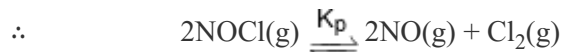
$$\text{Let } P_{\text{NOCl}} = x \text{ atm} \quad P_{\text{N}_2} = (1 - x) \text{ atm}$$



$$x-y \quad y \quad y/2$$

$$x-y + y + y/2 + (1-x) = 1.2$$

$$\text{or } y = 0.4$$



$$\text{At equilibrium } x-0.4 \quad 0.4 \quad 0.2$$

Now pressure due to added Cl_2 only if no further reaction occurs

$$= 8.3 - 1.2 = 7.1$$

$\therefore$ Pressure due to total Cl_2 if no further reaction occurs = $7.1 + 0.2 = 7.3$



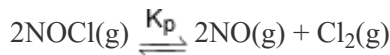
$$x-0.4 \quad 0.4 \quad 7.3 \quad P_{\text{total}_1} = 7.3 + x$$

$$x-0.4 + 2z \quad 0.4-2z \quad 7.3-z \quad P_{\text{total}_2} = 7.3 + x-z$$

$$p_{\text{total}_1} - p_{\text{total}_2} = 8.3 - 8.2$$

$$\text{or } 7.3 + x - (7.3 + x - z) = 0.1$$

$$\text{or } z = 0.1$$



$$\text{At equilibrium } x-0.2 \quad 0.2 \quad 7.2$$

$$\text{Now } (K_p)_{\text{at equilibrium 1}} = (K_p)_{\text{at equilibrium 2}}$$

$$\text{or } \frac{(0.4)^2 \times 0.2}{(x-0.4)^2} = \frac{(0.2)^2 \times (7.2)}{(x-0.2)^2}$$

$$\text{or } \frac{1}{(x-0.4)^2} = \frac{9}{(x-0.2)^2}$$

$$\text{or } \frac{1}{x-0.4} = \frac{3}{(x-0.2)}$$

$$\text{or } 3x - 1.2 = x - 0.2$$

$$\text{or } x = 0.5$$

$$\therefore K_p = \frac{(0.4)^2 \times 0.2}{(0.5-0.4)^2} = \frac{0.032}{0.01} = 3.2$$

5. **Ans (0.50)**

$$x \Rightarrow \text{(i), (iv)}$$

$$y \Rightarrow \text{(ii), (iii), (v), (vi)}$$

6. **Ans (1.33)**

$$\text{V}^{+4} \rightarrow \text{Blue}$$

$$\text{Cr}^{+3} \rightarrow \text{Violet}$$

$$\text{Fe}^{+3} \rightarrow \text{Yellow}$$

$$\text{Ni}^{+2} \rightarrow \text{Green}$$

$$\text{Cr}^{+2} \rightarrow \text{Blue}$$

$$\text{Co}^{+2} \rightarrow \text{Pink}$$

$$\text{Cu}^{+2} \rightarrow \text{Blue}$$

$$\text{Mn}^{+3} \rightarrow \text{Violet}$$

$$\text{V}^{+2} \rightarrow \text{Violet}$$

$$x = \text{V}^{+4}, \text{Cr}^{+2}, \text{Cu}^{+2}$$

$$y = \text{Fe}^{+3}$$

$$z = \text{Cr}^{+3}, \text{Mn}^{+3}, \text{V}^{+2}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C)**

$$\text{(i) } aabbcc = {}^9C_3 \frac{6!}{2!2!2!} = 7560$$

$$\text{(ii) } aaoooc = {}^9C_2 \left[\frac{6!}{2!2!2!} - \frac{5!}{2!2!} \right] = 2160$$

$$\text{(iii) } aaabbb \text{ or } aaaabb =$$

$${}^9C_2 \left(\frac{6!}{3!3!} + 2 \times \frac{6!}{4!2!} \right) = 1800$$

$$\text{(iv) } aaaooo \text{ or } aaoooo \text{ or } aaaaoo$$

$$= {}^9C_1 \left[\frac{6!}{3!3!} - \frac{5!}{3!2!} \right] + {}^9C_1 \left[\frac{6!}{4!2!} - \frac{5!}{4!1!} \right] + {}^9C_1 \left[\frac{6!}{4!2!} - \frac{5!}{3!2!} \right]$$

$$= 90 + 90 + 45$$

$$\text{(v) } aaaaaa = 9$$

$$\text{total} = 11754$$

2. **Ans (A,B)**

$$\int_0^1 e^x \frac{(x+1)^2 - 2x - 1}{(x+1)^2} dx$$

$$= (e-1) - \int e^x \left(\frac{2x+2}{(x+1)^2} - \frac{1}{(x+1)^2} \right) dx$$

$$(e-1) - \int_0^1 \frac{e^x}{x+1} dx - \int_0^1 e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right)$$

$$= e - 1 - a - \left[\frac{e^x}{x+1} \right]_0^1$$

$$\text{(B) } (e^x \log x)_1^2 - \int_1^2 \frac{e^x}{x} dx \text{ put } x = 1+t$$

$$(e^2 \log 2 - e \log 1) - \int_0^1 \frac{e^{t+1}}{t+1} dt$$

$$e^2 \log 2 - ea$$

3. Ans (B,D)

Here,

$$S_n = \frac{\sin(\pi - \theta)}{\sin \theta \cdot \sin 2\theta} + \frac{\sin(2\pi - 2\theta)}{\sin \theta \cdot \sin 3\theta} + \frac{\sin(3\pi - 3\theta)}{\sin \theta \cdot \sin 4\theta} + \dots$$

$$\therefore S_n = \frac{\sin \theta}{\sin \theta \cdot \sin 2\theta} + \frac{(-\sin 2\theta)}{\sin \theta \cdot \sin 3\theta} + \frac{\sin 3\theta}{\sin \theta \cdot \sin 4\theta} + \dots$$

$$S_n = \frac{\sin(2\theta - \theta)}{\sin \theta \cdot \sin 2\theta} - \frac{\sin(3\theta - \theta)}{\sin \theta \cdot \sin 3\theta} + \frac{\sin(4\theta - \theta)}{\sin \theta \cdot \sin 4\theta} \dots$$

$$S_n = (\cot \theta - \cot 2\theta) - (\cot \theta - \cot 3\theta) +$$

$$(\cot \theta - \cot 4\theta) - (\cot \theta - \cot 5\theta) + \dots$$

$$\Rightarrow S_{99} = (\cot \theta - \cot 2\theta) - (\cot \theta - \cot 3\theta) + \dots +$$

$$(\cot \theta - \cot 100\theta)$$

$$\therefore S_{99} = \cot \theta - \cot 2\theta + \cot 3\theta \dots - \cot 100\theta$$

$$\Rightarrow S_{100} = (\cot \theta - \cot 2\theta) - (\cot \theta - \cot 3\theta) + \dots -$$

$$(\cot \theta - \cot 101\theta)$$

$$= -\cot 2\theta + \cot 3\theta - \cot 4\theta + \dots + \cot 101\theta$$

4. Ans (A,B,C)

$$2\alpha_{r+2} = \alpha_r + \alpha_{r+1}$$

$$\Rightarrow 2(\alpha_{r+2} - \alpha_{r+1}) = \alpha_r - \alpha_{r+1}$$

$$\Rightarrow \alpha_{r+2} - \alpha_{r+1} = -\frac{1}{2}(\alpha_{r+1} - \alpha_r)$$

$$\Rightarrow \alpha_{10} - \alpha_9 = -\frac{1}{2}(\alpha_9 - \alpha_8)$$

$$= \frac{1}{4}(\alpha_8 - \alpha_7) = \dots$$

$$= -\frac{1}{2^9}(\alpha_1 - \alpha_0) = \frac{-16}{2^9} = \frac{-1}{32}$$

As $\alpha_0 - \alpha_1, \alpha_1 - \alpha_2, \alpha_2 - \alpha_3$, are in G.P.

$\Rightarrow \alpha_0 - \alpha_2, 2(\alpha_1 - \alpha_2), \alpha_1 - \alpha_3$ are in H.P.

(Adding middle term to all terms)

5. Ans (A,B,C,D)

$$P(A) = 1 - P(\bar{A}) = 1 - \left(\frac{5}{6}\right)^4$$

$$P(A \cap B) = 1 - P(\overline{A \cap B})$$

$$= 1 - P(\bar{A}) - P(\bar{B}) + P(\bar{A} \cap \bar{B})$$

$$= 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4$$

$$P(A \cap B | C \cap D) = \frac{P(A \cap B \cap C \cap D)}{P(C \cap D)}$$

$$= \frac{24/6^4}{1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4} = \frac{24}{302}$$

6. Ans (A,B,C)

$$\text{As } \lim_{x \rightarrow 1} (1 + \ln x \cdot \ln(k^2 + 25))^{\frac{1}{|x|-1}}$$

$$\lim_{x \rightarrow 1} (1 + \ln x \cdot \ln(k^2 + 25))^{\frac{1}{x-1}}$$

$$= e^{\lim_{x \rightarrow 1} \frac{\ln x \cdot \ln(k^2 + 25)}{x-1}} = e^{\ln(k^2 + 25)} = k^2 + 25$$

$$\therefore k^2 + 25 = k(2\sin^2 \alpha + 3\cos \beta + 5)$$

$$\Rightarrow 2\sin^2 \alpha + 3\cos \beta + 5 = k + \frac{25}{k}$$

$$\therefore \frac{k + \frac{25}{k}}{2} \geq \sqrt{k \cdot \frac{25}{k}}$$

$$\Rightarrow k + \frac{25}{k} \geq 10$$

But L.H.S. ≤ 10

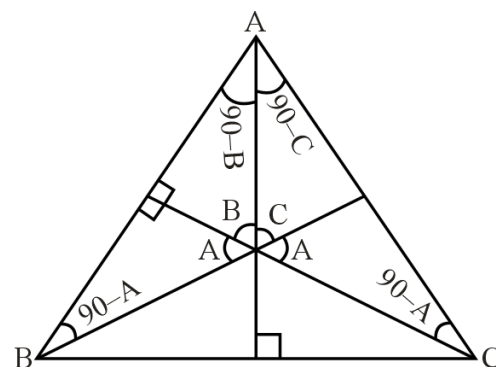
So both L.H.S. and R.H.S. should be equal to 10.

$$\Rightarrow \sin^2 \alpha = 1 \text{ and } \cos \beta = 1$$

$$\text{Hence, } \frac{\sin^{10} \alpha + \cos^5 \beta}{\sin^2 \alpha + \cos \beta} = \frac{1 + 1}{1 + 1} = 1$$

7. Ans (A,B,C)

$\therefore |z_1| = |z_2| = |z_3| \Rightarrow$ origin is the circumcentre of $\triangle ABC$



$\Rightarrow$ orthocentre of $\triangle ABC$ is $z_1 + z_2 + z_3$

Now $\angle AHB = A + B$ and $\angle AHC = A + C$

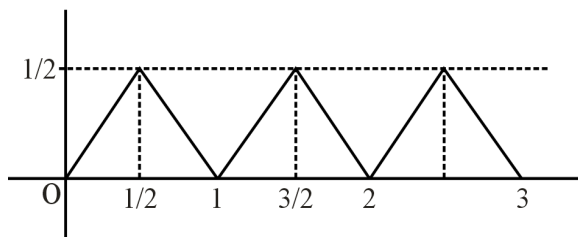
$$\Rightarrow \angle AHB = \angle AHC \quad (\because \angle B = \angle C)$$

$$\Rightarrow \arg\left(\frac{z_1 + z_2 + z_3 - z_2}{z_1 + z_2 + z_3 - z_1}\right) = \arg\left(\frac{z_1 + z_2 + z_3 - z_1}{z_1 + z_2 + z_3 - z_3}\right)$$

$$\Rightarrow \arg\left(\frac{z_1 + z_3}{z_2 + z_3}\right) = \arg\left(\frac{z_2 + z_3}{z_2 + z_1}\right)$$

$$\Rightarrow \arg\left(\frac{(z_1 + z_3) \cdot (z_1 + z_2)}{(z_2 + z_3)^2}\right) = 0$$

8. Ans (A,C,D)



PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (C)

(A) $e_1 < 1 < e_2$ $f(1) < 0 \Rightarrow a > 3$

(B) both roots > 1

(i) $D \geq 0$

(ii) $f(1) > 0 \Rightarrow a \in [2\sqrt{2}, 3)$

(iii) $a/2 > 1$

(C) $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \Rightarrow a = \pm 2\sqrt{2}$

(D) $1 < e_1 < \sqrt{2} \leq e_2$

$\Rightarrow f(\sqrt{2}) < 0$ and $f(1) > 0$

$\Rightarrow a \in (2\sqrt{2}, 3)$

10. Ans (B)

(A) $e_1 < 1 < e_2$ $f(1) < 0 \Rightarrow a > 3$

(B) both roots > 1

(i) $D \geq 0$

(ii) $f(1) > 0 \Rightarrow a \in [2\sqrt{2}, 3)$

(iii) $a/2 > 1$

(C) $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \Rightarrow a = \pm 2\sqrt{2}$

(D) $1 < e_1 < \sqrt{2} \leq e_2$

$\Rightarrow f(\sqrt{2}) < 0$ and $f(1) > 0$

$\Rightarrow a \in (2\sqrt{2}, 3)$

11. Ans (A)

(A) Given $\frac{1}{6} abc = 32 \Rightarrow abc = 192$ where $A(a, 0, 0)$,

$B(0, b, 0), C(0, 0, c)$

$(\alpha, \beta, \gamma) = \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right) \Rightarrow 64 \alpha\beta\gamma = 192$

$\Rightarrow$ locus is $xyz = 3$

(B) $(\alpha, \beta, \gamma) = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right) \Rightarrow 8\alpha\beta\gamma = 192$

$\Rightarrow 8\alpha\beta\gamma = 192 \Rightarrow xyz = 24$

(C) Equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

foot of perp $= (\alpha, \beta, \gamma) = \left(\frac{\frac{1}{a}}{\sum \frac{1}{a^2}}, \frac{\frac{1}{b}}{\sum \frac{1}{b^2}}, \frac{\frac{1}{c}}{\sum \frac{1}{c^2}}\right)$

$\Rightarrow \sum \frac{1}{a^2} = \frac{1}{a\alpha} = \frac{1}{b\beta} = \frac{1}{c\gamma}$

Let $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = t$

$\Rightarrow \frac{1}{a} = \alpha t, \frac{1}{b} = \beta t, \frac{1}{c} = \gamma t$

$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = (\alpha^2 + \beta^2 + \gamma^2)t^2$

$\Rightarrow \frac{1}{t} = \alpha^2 + \beta^2 + \gamma^2$

$\Rightarrow \frac{1}{t^3} = (\alpha^2 + \beta^2 + \gamma^2)^3$

$\Rightarrow abc \alpha\beta\gamma = (\alpha^2 + \beta^2 + \gamma^2)^3$

$\Rightarrow 192xyz = (x^2 + y^2 + z^2)^3$

(D) Let $P \equiv (x, y, z)$

$PA \perp PB \Rightarrow (x-a)(x) + y(y-b) + z(z) = 0$

$\Rightarrow x^2 + y^2 + z^2 = ax + by \dots(i)$

$PB \perp PC \Rightarrow x^2 + y(y-b) + z(z-c) = 0$

$\Rightarrow x^2 + y^2 + z^2 = by + cz \dots(ii)$

$PC \perp PA \Rightarrow x(x-a) + y^2 + z(z-c) = 0$

$\Rightarrow x^2 + y^2 + z^2 = ax + cz \dots(iii)$

$\Rightarrow ax + by = by + cz = ax + cz$

from (i) $2ax = x^2 + y^2 + z^2$

As $abc = 192 \Rightarrow 8(192)xyz = (x^2 + y^2 + z^2)^3$

12. Ans (C)

(A) Given $\frac{1}{6} abc = 32 \Rightarrow abc = 192$ where $A(a, 0, 0)$,
 $B(0, b, 0)$, $C(0, 0, c)$

$$(\alpha, \beta, \gamma) = \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4} \right) \Rightarrow 64 \alpha\beta\gamma = 192$$

$\Rightarrow$ locus is $xyz = 3$

(B) $(\alpha, \beta, \gamma) = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2} \right) \Rightarrow$ As $abc = 192$

$\Rightarrow 8\alpha\beta\gamma = 192 \Rightarrow xyz = 24$

(C) Equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

foot of perp = (α, β, γ)

$$= \left(\frac{\frac{1}{a}}{\sum \frac{1}{a^2}}, \frac{\frac{1}{b}}{\sum \frac{1}{b^2}}, \frac{\frac{1}{c}}{\sum \frac{1}{c^2}} \right)$$

$$\Rightarrow \sum \frac{1}{a^2} = \frac{1}{a\alpha} = \frac{1}{b\beta} = \frac{1}{c\gamma}$$

$$\text{Let } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = t$$

$$\Rightarrow \frac{1}{a} = \alpha t, \frac{1}{b} = \beta t, \frac{1}{c} = \gamma t$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = (\alpha^2 + \beta^2 + \gamma^2)t^2$$

$$\Rightarrow \frac{1}{t} = \alpha^2 + \beta^2 + \gamma^2$$

$$\Rightarrow \frac{1}{t^3} = (\alpha^2 + \beta^2 + \gamma^2)^3$$

$$\Rightarrow abc \alpha\beta\gamma = (\alpha^2 + \beta^2 + \gamma^2)^3$$

$$\Rightarrow 192xyz = (x^2 + y^2 + z^2)^3$$

(D) Let $P \equiv (x, y, z)$

$$PA \perp PB \Rightarrow (x-a)(x) + y(y-b) + z(z) = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = ax + by \quad \dots(i)$$

$$PB \perp PC \Rightarrow x^2 + y(y-b) + z(z-c) = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = by + cz \quad \dots(ii)$$

$$PC \perp PA \Rightarrow x(x-a) + y^2 + z(z-c) = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = ax + cz \quad \dots(iii)$$

$$\Rightarrow ax + by = by + cz = ax + cz$$

$$\text{from (i) } 2ax = x^2 + y^2 + z^2$$

$$\text{As } abc = 192 \Rightarrow 8(192)xyz = (x^2 + y^2 + z^2)^3$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (8.00)

Let the middle point of chord be $(t^2, 2t)$. Mid point of chord must lie inside the ellipse

$$\Rightarrow t^4 + 8t^2 - 1 < 0$$

$$\Rightarrow t^2 \in (0, -4 + \sqrt{17}) \quad \dots(1)$$

Also equation of the chord is $T = S_1$

$$\Rightarrow t^2x + 4ty = t^4 + 8t^2$$

Let this passes through $(a, 0)$

$$\Rightarrow at^2 = t^4 + 8t^2$$

$$\Rightarrow t^4 + (8 - a)t^2 = 0$$

$$\Rightarrow t^2 = 0 \text{ or } t^2 = a - 8$$

$$\Rightarrow a = t^2 + 8$$

$$\Rightarrow a \in (8, 4 + \sqrt{17})$$

2. Ans (2.00)

$$\text{Here } 3^{|kx-1|} = \left(\frac{1}{27} \right)^{\frac{-k}{3}[x]}$$

$$\therefore |kx - 1| = k[x].$$

If $x < 0$ the positive = negative. If $0 \leq x < 1/k$

then positive = 0. If $x \geq 1/k$ then $kx - 1 = k[x]$

$$\Rightarrow \{x\} = \frac{1}{k} \quad \dots(1)$$

$$\text{Also } x^2 + k^2 - 2kx = k$$

$$\therefore (x - k)^2 = k \quad \dots(2)$$

$$\text{Now, } S_n = \sum_{r=0}^n \left(\left(\frac{x}{k} - 1 \right)^{2r} + \frac{r}{k} \right) = \sum_{r=0}^n \left(\frac{(x-k)^2}{k^2} + \frac{r}{k} \right)$$

$$\therefore S_n = \sum_{r=0}^n \left(\left(\frac{1}{k} \right)^r + \frac{r}{k} \right)$$

$$\therefore S_{20} = \frac{k^{21} - 1}{k^{21} - k^{20}} + \frac{20 \cdot 21}{2k}$$

3. Ans (10.00)

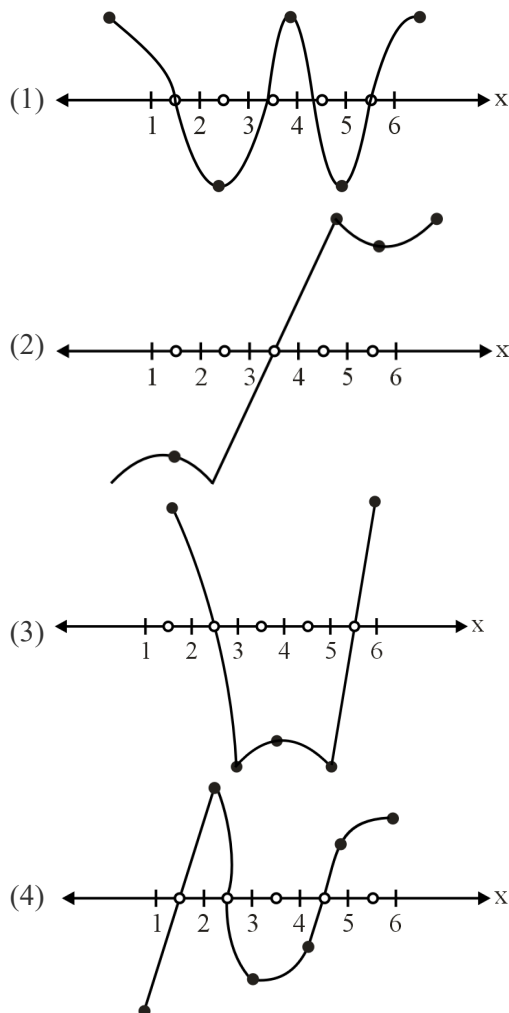
$$f(1) f(2) f(3) f(4) f(5) f(6) < 0$$

$$f(2) f(4) f(6) < 0$$

$$f(3) f(6) < 0$$

$f(6) > 0$, $f(3) < 0$, $f(2)$, $f(4)$ will be opposite sign. $f(1)$, $f(5)$ will be opposite sign.

Following 4 possibilities are as



Number of possible roots 1, 2, 3, 4

$$A = \{1, 2, 3, 4\}$$

$$\text{Hence, } \Sigma n(A) = 10$$

4. Ans (8.00)

For first row $a_{11}, a_{12}, a_{13}, \dots, a_{1(n-1)}$ can be either 1 or $-1 \Rightarrow 2^{n-1}$ ways

For a_{1n} only one way depending upon the product $(a_{11} \cdot a_{12} \cdot a_{13} \cdot \dots \cdot a_{1(n-1)})$

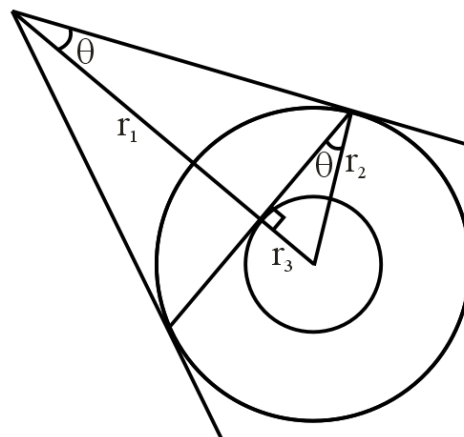
$\therefore$ For R_1 we have 2^{n-1} ways

Similarly for $R_2, R_3, \dots, R_{n-1}$, we have 2^{n-1} ways

$\therefore$ For R_n only one way

$$\therefore \text{Total ways } (2^{n-1})^{n-1} = 2^{(n-1)^2}$$

5. Ans (20.00)



$$\sin \theta = \frac{r_2}{r_1} = \frac{r_3}{r_2} = \dots$$

$\Rightarrow r_1, r_2, r_3, \dots$ are in G.P. with common ratio

$$= \sin \frac{\pi}{6} = \frac{1}{2}$$

$$\Rightarrow \text{Sum of all } r_i = \frac{r_1}{1 - 1/2} = 2r_1 = 20$$

6. Ans (2.00)

$$x - y + 2 = 0 \text{ \& } -7x + y + 8 = 0$$

$$\therefore a_1 a_2 + b_1 b_2 < 0$$

$$\therefore \text{Acute angle bisector is } \frac{x-y+2}{\sqrt{2}} = + \frac{-7x+y+8}{5\sqrt{2}}$$

$$\Rightarrow 6x - 3y + 1 = 0$$

$$\Rightarrow \text{Area} = \frac{1}{36}$$